

# Othmane Echchabi

[othmaneechc.github.io/](https://othmaneechc.github.io/)

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## EDUCATION

### McGill University & Mila - Quebec AI Institute

*Master's of Science in Computer Science*

Montréal, QC

*Aug. 2025 – Present*

### Duke University / Duke Kunshan University

*Bachelor's of Science in Data Science*

Durham, NC / Suzhou, China

*Aug. 2021 – May 2025*

- Thesis: *Monitoring Spartina Alterniflora Using Self-Supervised Learning*
- Study Abroad: Universidad Carlos III de Madrid - Spring 2022.

## PROGRAMMING SKILLS & LANGUAGES

**Frameworks/Developer Tools:** React, Node.js, Flask, FastAPI, Git, Docker, VS Code.

**Languages/Libraries:** Python, Java, C/C++, SQL (Postgres), JavaScript, R, PyTorch, TensorFlow, Scikit-Learn.

**Spoken Languages:** Arabic (native); French (fluent); English (fluent); Chinese (intermediate); Spanish (conversational)

## EXPERIENCE

### Machine Learning Research Intern

*University Mohammed VI Polytechnic*

Jul. 2025 – Present

*Benquerir, Morocco*

- Unified a 10-yr global crop-yield to a fertilizer dataset (N/P/K) across 150+ countries.
- Shipped transformer field-recs and an agentic LLM + retrieval copilot (RAG) with 2-second lookup latency.

### Undergraduate Research Assistant

*Duke Kunshan University*

Jun. 2024 – Aug. 2025

*Suzhou, China*

- Built a 300k-image, ground-truthed satellite dataset across 50/54 African countries; trained DINO (ViT backbone) model to 97% accuracy (*supervised by Dr. Ka Leung Lam*).
- *SpeedTransformer*: SOTA 96% Top-1 on Geolife transport-mode prediction (+3%); launched *CarbonClever*, a social app turning mobility insights into personal CO<sub>2</sub> reductions (*supervised by Dr. Charles Chang*).

### Undergraduate Research Assistant

*University of Missouri*

Jan. 2025 – Jun. 2025

*Columbia, MO*

- Built a Perceiver IO-based foundation model and a VAE simulator for wildfire spread from satellite + climate data (long-range spatiotemporal dynamics, scenario analysis) (*Supervised by Dr. Jordan Malof*).

### Undergraduate Research Fellow

*Duke University*

Jun. 2023 – Aug. 2024

*Durham, NC*

- Mapped invasive salt-marsh species with ViTs on Sentinel-2/Landsat; presented landscape-scale outputs at [Duke Climate+ 2024](#) to inform conservation policy (*Supervised by Dr. Wenhong Li & Dr. Ding Ma*).

### Data Analyst Intern

*XPerlean*

Jul. 2022 – Aug. 2022

*Saint-Quentin, France*

- Designed a data engineering pipeline to extract insightful metrics from industrial/factory collected data.
- Used R-CNN & YOLO for ceramic defect detection, raised accuracy by 90%+; boosted production speed by 10%.

### Data Analyst Intern

*Atos Morocco*

Oct. 2022 – Nov. 2022

*Rabat, Morocco*

- Built real-time HR dashboards used by leadership using Microsoft Power BI; scaled across African branches.

## PUBLICATIONS & PREPRINTS

**O. Echchabi**, et al. *Tracking Progress Towards SDG 6 Using Satellite Imagery*. arXiv:2411.19093.

**O. Echchabi**, N. Talty. *Football AI Tracker: End-to-End Player Tracking from Low-Quality Recording*.

Z. Yuandong\*, **O. Echchabi**\*, et al. *SpeedTransformer: Predicting Human Mobility with Dense GPS and Transformers* (Under Review).

## GRANTS & AWARDS

McGill CS Graduate Award \$30k/yr (2025); McGill Registration Bonus \$5k (2025); DKU Experiential Fellowship \$1k (2024); Signature Work Grant \$1.5k (2023); Innovation Incubator \$3k (2023)